

Exercise 02: Causal Self-Attention

The same attention calculation as Exercise 01, with one new object: the causal mask.

Setup

We use the same sentence, embeddings, and projection matrices as Exercise 01:

“I like cats”.

The three token rows are I , like, and cats. Here $N = 3$, $d_{\text{model}} = 2$, $d_k = 2$, and $d_v = 2$.

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

$$W_Q = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

The causal attention equation is

$$O_{\text{causal}} = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + M \right) V.$$

The mask M is added to the scaled logits before softmax.

1. Compute $Q = XW_Q$

Since $W_Q = I_2$,

$$Q = XW_Q = XI_2 = X.$$

Therefore

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

2. Compute $K = XW_K$

$$k_I = [1 \quad 0] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [1 \quad 1],$$

$$k_{\text{like}} = [0 \quad 1] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [0 \quad 1],$$

$$k_{\text{cats}} = [1 \quad 1] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [1 \quad 2].$$

Thus

$$K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

3. Compute $V = XW_V$

$$v_I = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

$$v_{\text{like}} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \end{bmatrix},$$

$$v_{\text{cats}} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \end{bmatrix}.$$

Thus

$$V = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

4. Compute $K^\top$

$$K^\top = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$$

5. Compute Raw Scores $S = QK^\top$

The entries of S are query-key dot products.

For $q_I = [1, 0]$:

$$q_I \cdot k_I = (1)(1) + (0)(1) = 1,$$

$$q_I \cdot k_{\text{like}} = (1)(0) + (0)(1) = 0,$$

$$q_I \cdot k_{\text{cats}} = (1)(1) + (0)(2) = 1,$$

For $q_{\text{like}} = [0, 1]$:

$$q_{\text{like}} \cdot k_I = (0)(1) + (1)(1) = 1,$$

$$q_{\text{like}} \cdot k_{\text{like}} = (0)(0) + (1)(1) = 1,$$

$$q_{\text{like}} \cdot k_{\text{cats}} = (0)(1) + (1)(2) = 2,$$

For $q_{\text{cats}} = [1, 1]$:

$$q_{\text{cats}} \cdot k_I = (1)(1) + (1)(1) = 2,$$

$$q_{\text{cats}} \cdot k_{\text{like}} = (1)(0) + (1)(1) = 1,$$

$$q_{\text{cats}} \cdot k_{\text{cats}} = (1)(1) + (1)(2) = 3.$$

Therefore

$$S = QK^\top = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

6. Compute Scaled Scores

Since $d_k = 2$,

$$\sqrt{d_k} = \sqrt{2} \approx 1.4142.$$

$$S_{\text{scaled}} = \frac{QK^\top}{\sqrt{d_k}} = \begin{bmatrix} 0.7071 & 0.0000 & 0.7071 \\ 0.7071 & 0.7071 & 1.4142 \\ 1.4142 & 0.7071 & 2.1213 \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

7. Construct the Causal Mask

For this square $N \times N$ self-attention example, rows are querying/current tokens and columns are tokens being attended to. A column index $j > i$ means the querying token would look into the future.

The causal mask is

$$M_{ij} = \begin{cases} 0, & j \leq i, \\ -\infty, & j > i. \end{cases}$$

For three tokens,

$$M = \begin{bmatrix} 0 & -\infty & -\infty \\ 0 & 0 & -\infty \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

The forbidden future-token connections are:

$$\begin{aligned} M_{1,2} : & \quad I \rightarrow \text{like}, & \text{future}, \\ M_{1,3} : & \quad I \rightarrow \text{cats}, & \text{future}, \\ M_{2,3} : & \quad \text{like} \rightarrow \text{cats}, & \text{future}. \end{aligned}$$

8. Compute Masked Scores

Add the mask to the scaled logits:

$$S_{\text{masked}} = S_{\text{scaled}} + M.$$

$$S_{\text{masked}} = \begin{bmatrix} 0.7071 & -\infty & -\infty \\ 0.7071 & 0.7071 & -\infty \\ 1.4142 & 0.7071 & 2.1213 \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

Future-token logits have become $-\infty$ before softmax.

9. Apply Row-Wise Softmax

Softmax is applied independently to each row:

$$\text{softmax}(z)_j = \frac{e^{z_j}}{\sum_{\ell} e^{z_{\ell}}}.$$

The key effect of the mask is

$$e^{-\infty} = 0.$$

So masked future positions receive exactly zero probability.

First row

$$z_I = [0.7071 \quad -\infty \quad -\infty] .$$

$$e^{0.7071} \approx 2.0281, \quad e^{-\infty} = 0.$$

$$Z_I = 2.0281 + 0 + 0 = 2.0281.$$

$$a_I = \begin{bmatrix} \frac{2.0281}{2.0281} & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 \end{bmatrix} .$$

Second row

$$z_{\text{like}} = [0.7071 \quad 0.7071 \quad -\infty] .$$

$$e^{0.7071} \approx 2.0281, \quad e^{-\infty} = 0.$$

$$Z_{\text{like}} = 2.0281 + 2.0281 + 0 = 4.0562.$$

$$a_{\text{like}} = \begin{bmatrix} \frac{2.0281}{4.0562} & \frac{2.0281}{4.0562} & 0 \end{bmatrix} = \begin{bmatrix} 0.5000 & 0.5000 & 0.0000 \end{bmatrix} .$$

Third row

Nothing is masked in the third row, because cats is the final token. This is the same row as the cats row in Exercise 01.

$$z_{\text{cats}} = [1.4142 \quad 0.7071 \quad 2.1213] .$$

$$e^{1.4142} \approx 4.1132,$$

$$e^{0.7071} \approx 2.0281,$$

$$e^{2.1213} \approx 8.3420.$$

$$Z_{\text{cats}} \approx 4.1132 + 2.0281 + 8.3420 = 14.4833.$$

$$a_{\text{cats}} = [0.2840 \quad 0.1400 \quad 0.5760] .$$

Therefore

$$A_{\text{causal}} = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 \\ 0.5000 & 0.5000 & 0.0000 \\ 0.2840 & 0.1400 & 0.5760 \end{bmatrix} \in \mathbb{R}^{3 \times 3} .$$

10. Verify Row Sums and Masked Zeros

The row sums are

$$\begin{aligned} 1.0000 + 0.0000 + 0.0000 &= 1.0000, \\ 0.5000 + 0.5000 + 0.0000 &= 1.0000, \\ 0.2840 + 0.1400 + 0.5760 &\approx 1.0000. \end{aligned}$$

Every future-token entry is zero:

$$A_{\text{causal},1,2} = 0, \quad A_{\text{causal},1,3} = 0, \quad A_{\text{causal},2,3} = 0.$$

11. Compute $O_{\text{causal}} = A_{\text{causal}}V$

The value vectors are

$$v_I = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad v_{\text{like}} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad v_{\text{cats}} = \begin{bmatrix} 2 & 1 \end{bmatrix}.$$

For I :

$$o_I = 1.0000v_I = \begin{bmatrix} 1.0000 & 0.0000 \end{bmatrix}.$$

For like:

$$\begin{aligned} o_{\text{like}} &= 0.5000v_I + 0.5000v_{\text{like}}, \\ &= 0.5000 \begin{bmatrix} 1 & 0 \end{bmatrix} + 0.5000 \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1.0000 & 0.5000 \end{bmatrix}. \end{aligned}$$

For cats:

$$\begin{aligned} o_{\text{cats}} &= 0.2840v_I + 0.1400v_{\text{like}} + 0.5760v_{\text{cats}}, \\ o_{\text{cats},1} &= 0.2840(1) + 0.1400(1) + 0.5760(2) \approx 1.5760, \\ o_{\text{cats},2} &= 0.2840(0) + 0.1400(1) + 0.5760(1) \approx 0.7160. \end{aligned}$$

Therefore

$$O_{\text{causal}} = \begin{bmatrix} 1.0000 & 0.0000 \\ 1.0000 & 0.5000 \\ 1.5760 & 0.7160 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

12. Compare with Exercise 01

The unmasked attention matrix from Exercise 01 was

$$A_{\text{unmasked}} = \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \\ 0.2483 & 0.2483 & 0.5035 \\ 0.2840 & 0.1400 & 0.5760 \end{bmatrix}.$$

The causal attention matrix is

$$A_{\text{causal}} = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 \\ 0.5000 & 0.5000 & 0.0000 \\ 0.2840 & 0.1400 & 0.5760 \end{bmatrix}.$$

The unmasked output from Exercise 01 was

$$O_{\text{unmasked}} = \begin{bmatrix} 1.4011 & 0.5989 \\ 1.5035 & 0.7517 \\ 1.5760 & 0.7160 \end{bmatrix}.$$

The causal output is

$$O_{\text{causal}} = \begin{bmatrix} 1.0000 & 0.0000 \\ 1.0000 & 0.5000 \\ 1.5760 & 0.7160 \end{bmatrix}.$$

The central observation is:

I	changes,	because it may attend only to $[I]$,
like	changes,	because it may attend only to $[I, \text{like}]$,
cats	does not change,	because it may attend to $[I, \text{like}, \text{cats}]$.

Since cats is already the final token, there are no future positions to mask in its row.

Final Question

The attention row for cats is unchanged because the third row of the causal mask contains no $-\infty$ entries:

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}.$$

The token cats may attend to I , like, and cats itself. By contrast, I and like both have future positions removed before softmax, so their probability distributions change.

Bonus Question

The causal mask must be applied before softmax because softmax normalizes by a denominator. If probabilities are zeroed afterward, the denominator has already included the forbidden future tokens.

For example, the unmasked first attention row from Exercise 01 was

$$\begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \end{bmatrix}.$$

If we softmax first and then simply zero future positions, we get

$$\begin{bmatrix} 0.4011 & 0 & 0 \end{bmatrix}, \quad 0.4011 + 0 + 0 = 0.4011.$$

That is no longer a normalized attention row.

By masking the logits first,

$$\begin{bmatrix} 0.7071 & 0.0000 & 0.7071 \end{bmatrix} \longrightarrow \begin{bmatrix} 0.7071 & -\infty & -\infty \end{bmatrix},$$

so softmax normalizes only over legal positions:

$$\text{softmax} \begin{bmatrix} 0.7071 & -\infty & -\infty \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

The row still sums to 1.

Pipeline Summary

The full causal attention pipeline is:

$$\begin{aligned} Q &= XW_Q \\ X &\longrightarrow K = XW_K \longrightarrow S = QK^\top \longrightarrow S_{\text{scaled}} = \frac{QK^\top}{\sqrt{d_k}} \\ V &= XW_V \\ &\longrightarrow S_{\text{masked}} = S_{\text{scaled}} + M \longrightarrow A_{\text{causal}} = \text{softmax}(S_{\text{masked}}) \longrightarrow O_{\text{causal}} = A_{\text{causal}}V. \end{aligned}$$

Collapsed into the single causal attention equation:

$$O_{\text{causal}} = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + M \right) V.$$

Numerical Verification

The companion file `answer_key.py` verifies the manually derived matrices using basic PyTorch tensor operations.